

AI Notes for XML/SOAP → REST/API Bridge

Project Documentation and Sample AI Notes

Project Overview

The XML/SOAP to REST/API Bridge converts legacy SOAP services into modern REST APIs. AI Notes automatically analyze requests, responses, mappings, errors, and system health.

Example 1: SOAP Request Analysis

Operation: GetUser. Purpose: Retrieve customer information. Parameters: UserId, AuthToken.
REST Equivalent: GET /api/users/{id}. Status: Valid Request.

Example 2: REST Request Analysis

REST Endpoint: GET /api/users. Mapped SOAP Operation: GetUser. SOAP Translation Generated Successfully.

Example 3: SOAP Fault Analysis

Fault Code AUTH_001. Root Cause: Invalid Authentication. Suggested Resolution: Generate a new token and retry.

Screen 1: WSDL Analyzer

Parses WSDL files, identifies operations, complex types, and generates REST endpoints automatically.

Screen 2: AI Field Mapper

Maps SOAP XML fields to REST JSON fields and calculates mapping confidence.

Screen 3: API Sandbox

Allows testing REST requests and viewing generated SOAP XML in real time.

Screen 4: Error Intelligence

Explains SOAP faults and validation errors in plain English with suggested fixes.

Screen 5: API Monitoring

Tracks request volume, success rate, error trends, and bridge health metrics.

Screen 6: Security Analyzer

Identifies authentication issues, exposed endpoints, and missing security configurations.

Screen 7: Source Code Generator

Generates Spring Boot controllers, SOAP clients, JAXB models, security configuration, and Swagger documentation.

Screen 8: Audit Logs

Stores complete request history, SOAP operations, REST endpoints, timestamps, and execution results.

Benefits

Simplifies XML payloads, improves debugging, accelerates integration, enhances observability, and reduces maintenance effort.

Conclusion

AI Notes provide intelligent insights for SOAP-to-REST transformation, making legacy integration easier for developers and enterprises.